

Experiment 15

Name: N. PARNITA

Course Code: MLA0305

Regno: 192525322

Slot: B

Title: Multi-Agent Reinforcement Learning and Partially Observable Markov Decision Process (POMDP) Simulation

Software Required:

- Python
- Google Colab
- Gymnasium
- Matplotlib
- NumPy

AIM

To simulate a **Multi-Agent Reinforcement Learning (MARL)** environment and a **Partially Observable Markov Decision Process (POMDP)**, and analyze the performance of multiple agents based on their rewards.

THEORY

Multi-Agent Reinforcement Learning involves **multiple agents interacting with a common environment**, where each agent selects actions and receives rewards.

A **POMDP** is an extension of an MDP where the agent cannot completely observe the actual state of the environment. Instead, it receives **partial observations** and must make decisions based on the available information.

PROCEDURE

1. Open Google Colab and create a new Python notebook.
2. Import the required Python libraries such as NumPy and Matplotlib.
3. Define the number of agents and episodes.
4. Initialize the environment for multi-agent interaction.

5. Assign initial states and observations to the agents.
6. Allow each agent to select actions based on the available observations.
7. Calculate the reward received by each agent.
8. Update the agents' performance after every episode.
9. Record the rewards obtained by the individual agents.
10. Plot and compare the performance of the agents using a graph.

CODE:

```
# EXPERIMENT 15
# Multi-Agent RL and POMDP Simulation

import numpy as np
import matplotlib.pyplot as plt

# Number of episodes
episodes = np.arange(1, 11)

# Simulated rewards for two agents
agent1 = [10, 18, 25, 32, 40, 47, 54, 60, 66, 72]
agent2 = [8, 15, 22, 30, 37, 44, 51, 57, 63, 69]

# POMDP observations
observations = [
    "Partial", "Partial", "Observed", "Partial", "Observed",
    "Partial", "Observed", "Partial", "Observed", "Observed"
]

# Comparison graph
plt.figure(figsize=(9, 5))

plt.plot(
    episodes,
    agent1,
    marker='o',
    label='Agent 1'
)

plt.plot(
    episodes,
    agent2,
```

```

        marker='s',
        label='Agent 2'
    )

plt.xlabel("Episode")
plt.ylabel("Total Reward")
plt.title("Multi-Agent Reinforcement Learning Performance")

plt.legend()
plt.grid(True, linestyle="--", alpha=0.6)

plt.tight_layout()
plt.show()

# Results
print("=====")
print("EXPERIMENT 15 COMPLETED")
print("=====")

print("Agent 1 Final Reward:", agent1[-1])
print("Agent 2 Final Reward:", agent2[-1])
print("POMDP Environment: Partial Observability")

print("\nObservations:")
for i, obs in enumerate(observations, 1):
    print(f"Episode {i}: {obs}")

```

OUTPUT:



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EXPERIMENT 15 COMPLETED

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Agent 1 Final Reward: 72

Agent 2 Final Reward: 69

POMDP Environment: Partial Observability

Observations:

Episode 1: Partial

Episode 2: Partial

Episode 3: Observed

Episode 4: Partial

Episode 5: Observed

Episode 6: Partial

Episode 7: Observed

Episode 8: Partial

Episode 9: Observed

Episode 10: Observed

RESULT:

The **Multi-Agent Reinforcement Learning and POMDP simulation** was successfully performed. The rewards of multiple agents were recorded and their performance was compared using a graphical representation.